1. ABSTRACT

In this project I worked on three experiments that are meant to explore image filtering techniques and their different effects by using morphological operators and spatial filters to denoise and enhance digital images. Using MATLAB, I conducted three experiments. The first experiment involved making a cleanup on a binary number using morphological opening to remove isolated noise specks outside the digits, followed by morphological closing to fill dark holes inside the digit strokes, together forming a clean image. The second experiment consisted of using two methods for cleaning a fingerprint image. The morphological method cleaned up the ridges while the median filter smoothed out the noise without losing details from the original image. The third experiment dealt with sharpening by using the Laplacian kernel method bring out details on an image of the moon.

The results revealed that each technique has a unique use case. Morphological operators such as opening and closing worked great for manipulating defined shapes, whereas median filters where effective on fixing impulse noise by working directly on grayscale data. The Laplacian sharpening helps to bring out the details on the edges, which did come at a cost of adding to the noise.

2. TEST RESULTS

2.1 Part 1 – Identify and Apply Morphological Operators

This experiment uses morphological opening followed by closing to clean a binary image. Opening removes isolated noise specks outside the digit strokes, while closing fills dark holes left inside them. This combination works very effectively on black and white images with already defined shapes.

Figure 1: Input image: Original white "2" on black binary image with noise.

Figure 2: Output image: Erosion shrinks bright regions, thinning the digit outline and removing small noise specks.

Figure 3: Input image: Original white "2" on black binary image with noise.

Figure 4: Output image: Open & Close removes external noise specks and fills dark holes inside the digits.

2.2 Part 2 – Fingerprint Denoising: Morphological vs. Median

Two filtering methods are used on this experiment to denoise a grayscale fingerprint image. The first method is morphological opening and closing, using an structuring element with the radius 1. This is to remove bright noise and seal dark gaps without losing fine detail. The second method is using a 3x3 median filter to reduce noise without affecting image sharpness. When choosing one over the other, it is best to determine if eliminating the most noise is more important, which morphological opening/closing is better at, or if preserving sharpness while removing most of the noise is priority, which is where 3×3 median filter excels.

Figure 7: Original fingerprint image displaying noise.

Figure 8: Morphological opening/closing with a radius-1 disk filtered: noise removed, small gaps sealed.

Figure 9: 3×3 median filtered: impulse noise suppressed, edges preserved.

2.3 Part 3 – Laplacian Enhancement

This experiment is using Laplacian sharpening to bring out the details of a fuzzy grayscale image of the moon. This method convolves the image with a negative-center Laplacian kernel which highlights the edges, then subtracts this response from the original image, resulting in better definition of the craters and surface texture. This works because the negative-centered kernel produces negative value at the edges, and subtracting those negative values adds contrast to the original intensity at those locations.

Figure 10: Original grayscale moon photo.

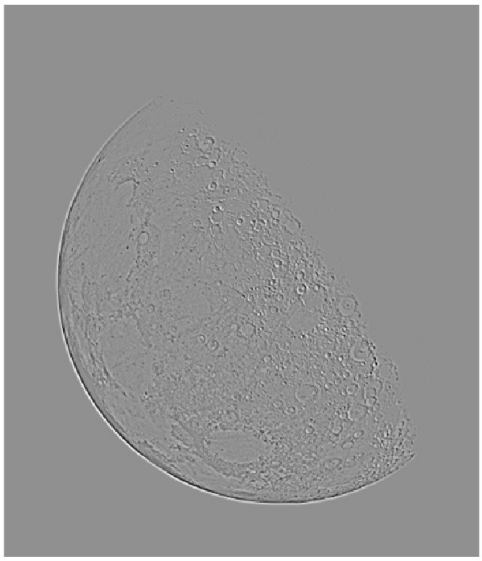

Figure 11: Laplacian response rescaled: highlights edges for crater rims and shadows.

Figure 12: Sharpened moon image: original minus Laplacian, showing crisper details.

3. DISCUSSION

These experiments gave me a better understanding of the effects of morphological tools and spatial filters. The right choice really depends on the image and what needs to be cleaned. Opening and closing were good methods for working with defined shapes in binary black and white images, while the median filter was better suited for image with grayscale data. The Laplacian sharpening works great for defining edges, but not meant for cleaning up noise. For further experimentation I would like to combine different sharpening techniques for better results, maybe come up with a solution for reducing noise while still bringing out details, especially when using the Laplacian technique.

4. CODES

```
% Name: Mauricio Gonzalez
% KSU ID: 001189051
% Assignment 3
clear all;
close all;
clc;

% CODE FOR EXPERIMENT 1 - Identify and Apply Morphological Operators
```

```
% Load image
srcMorph = imread('morphology.png');

% Ensure binary representation
if size(srcMorph, 3) == 3
   srcMorph = rgb2gray(srcMorph);
end
bwInput = imbinarize(srcMorph);

% Display the original noisy binary image
figure;
imshow(bwInput);

% Compact disk structuring element (radius 2)
seDisk = strel('disk', 2);

% Erosion using the disk structuring element
erodedImg = imerode(bwInput, seDisk);

% Display result
figure;
imshow(erodedImg);

% Opening strips away isolated bright specks/noise
openedImg = imopen(bwInput, seDisk);

% Closing seals any residual dark holes inside digits
dilatedImg = imclose(openedImg, seDisk);

% Display result
figure;
imshow(dilatedImg);

% CODE FOR EXPERIMENT 2 - Fingerprint Denoising: Morphological vs. Median

% Load image: converts to grayscale if color
% binarizes it into a black-and-white image for morphological tasks
fpRaw = imread('fingerprint_BW.png');
if size(fpRaw, 3) == 3
   fpRaw = rgb2gray(fpRaw);
end
fpBinary = imbinarize(fpRaw);

% Display original fingerprint
figure;
imshow(fpRaw);
```

```
% Morphological denoising open then close with a disk SE of radius 1
seDisk = strel('disk', 1);
fpMorphClean = imopen(fpBinary, seDisk);     % remove isolated bright specks
fpMorphClean = imclose(fpMorphClean, seDisk); % seal ridge breaks

% Display result
figure;
imshow(fpMorphClean);

% Median filter with a 3x3 neighborhood window
fpMedianClean = medfilt2(fpRaw, [3 3]);

% Display result
figure;
imshow(fpMedianClean);

% CODE FOR EXPERIMENT 3 - Laplacian Enhancement

% Load image: converts to grayscale if it's color, then converts to double precision
moonOrig = imread('moon.jpg');
if size(moonOrig, 3) == 3
    moonGray = rgb2gray(moonOrig);
else
    moonGray = moonOrig;
end
moonDbl = im2double(moonGray);

% Display original moon image
figure;
imshow(moonGray);

% Standard Laplacian kernel with negative center
lapMask = [0  1  0;
          1 -4  1;
          0  1  0];

% Compute the Laplacian response using border replication
lapResponse = imfilter(moonDbl, lapMask, 'replicate');

% Display result
figure;
imshow(im2uint8(mat2gray(lapResponse)));

% Sharpen by subtracting the Laplacian using negative-center convention
moonSharpened = moonDbl - lapResponse;

% Rescale to valid [0,1] range and convert to uint8
moonSharpened = im2uint8(mat2gray(moonSharpened));
```

```
% Display result
figure;
imshow(moonSharpened);
```